CS24303 — THEORY (3.0 CR) | 10-PAGE MASTER REVISION | BIT MESRA | NEP SCHEME

Data Mining — 10-Page Master Quick Revision

High-Yield Formula Sheet, Schema Matrices & Top BIT Mesra PYQ Solutions

⚡ 10-Page Master Quick Revision — Data Mining Concepts & Techniques (CS24303)

📋 The Complete 46-Topic Syllabus Master Checklist

46 / 46 Syllabus Topics Complete

Module	Topic #	Topic Name	Status
M1: Introduction (14)	1	Introduction to Data Mining (KDD Process)	✅ Complete
	2	Relational Databases (Tables & Schemas)	✅ Complete
	3	Data Warehouses (SITN & OLAP) [UPLOADED PYQ]	✅ Complete
	4	Transactional Databases & Market Basket Data	✅ Complete
	5	Advanced Database Systems & Applications	✅ Complete
	6	Data Mining Functionalities (Characterization vs. Discrimination) [UPLOADED PYQ]	✅ Complete
	7	Classification of Data Mining Systems	✅ Complete
	8	Major Issues in Data Mining (Quality, Scalability) [UPLOADED PYQ]	✅ Complete
	9	Data Concepts (Objects & Collections)	✅ Complete
	10	Data Objects & Attribute Types (Nominal, Binary, Ordinal, Numeric)	✅ Complete
	11	Basic Statistical Descriptions (Five-Number Summary, Box Plot)	✅ Complete
	12	Data Visualization (Scatter Plots, Histograms)	✅ Complete
	13	Measuring Data Similarity (Cosine Similarity)	✅ Complete
	14	Measuring Data Dissimilarity (Euclidean & Manhattan Distances)	✅ Complete
M2: Preprocessing (6)	15	Data Cleaning (Missing Values & Smoothing)	✅ Complete
	16	Data Integration (Schema Matching & Redundancy)	✅ Complete
	17	Data Transformation (Min-Max & Z-Score) [UPLOADED PYQ]	✅ Complete
	18	Data Reduction (PCA, Numerosity, Sampling)	✅ Complete
	19	Data Discretization (Binning, Histograms)	✅ Complete
	20	Concept Hierarchy Generation (Nominal & Numeric) [UPLOADED PYQ]	✅ Complete
M3: Data Warehouse (9)	21	Basic Concepts of Data Warehouse & ETL	✅ Complete

Module	Topic #	Topic Name	Status
	22	Data Warehouse Modeling (Star vs. Snowflake) [UPLOADED PYQ]	✅ Complete
	23	Data Cube (Base to Apex Cuboids)	✅ Complete
	24	OLAP Operations (Roll-Up, Drill-Down, Slice, Dice, Pivot) [UPLOADED PYQ]	✅ Complete
	25	Data Warehouse Design & 4 Views [UPLOADED PYQ]	✅ Complete
	26	Data Warehouse Implementation & Indexing	✅ Complete
	27	Attribute-Oriented Induction (AOI Generalization)	✅ Complete
	28	Data Cube Computation (2^n Cuboids & Materialization) [UPLOADED PYQ]	✅ Complete
	29	Preliminary Warehouse Concepts (Fact vs. Dimension Tables)	✅ Complete
M4: Patterns (7)	30	Basic Concepts of Frequent Pattern Mining	✅ Complete
	31	Associations (Support & Confidence Formulas)	✅ Complete
	32	Correlations (Lift Metric Formulation) [UPLOADED PYQ]	✅ Complete
	33	Frequent Itemset Mining Methods Overview	✅ Complete
	34	The Apriori Algorithm (Join & Prune Steps) [UPLOADED PYQ]	✅ Complete
	35	The FP-Growth Algorithm (FP-Tree Construction) [UPLOADED PYQ]	✅ Complete
	36	Interesting Pattern Evaluation (S-C-L Triad)	✅ Complete
M5: Advanced (10)	37	Pattern Mining Road Map	✅ Complete
	38	Multilevel Pattern Mining	✅ Complete
	39	Multidimensional Pattern Mining	✅ Complete
	40	Constraint-Based Frequent Pattern Mining	✅ Complete
	41	Mining High-Dimensional Data & Colossal Patterns	✅ Complete

Module	Topic #	Topic Name	Status
	42	Mining Compressed Patterns (Closed vs. Maximal)	✅ Complete
	43	Mining Approximate Patterns	✅ Complete
	44	Pattern Exploration & Filtering	✅ Complete
	45	Pattern Applications in Real-World Domains	✅ Complete
	46	**PYQ Block:** Classification (Trees, Bayes, Backprop) & Clustering (K-Means, PAM, DBSCAN, BIRCH)	✅ Complete

🧠 DMCT Master Algorithm Memory Cues

Algorithm	Memory Cue & Core Execution Loop
Apriori	`Generate -> Join -> Prune -> Count` (Uses anti-monotonicity to prune infrequent itemset supersets).
FP-Growth	`Compress -> Grow -> Mine` (Builds compact FP-tree and mines conditional pattern bases without candidate generation).
K-Means	`Assign -> Average -> Repeat` (Assigns to nearest centroid, recalculates means, minimizes SSE).
PAM (K-Medoids)	`Medoid -> Assign -> Swap -> Cost -> Repeat` (Uses actual representative objects; robust to outliers).
DBSCAN	`Core -> Expand -> Border -> Noise` (Density-based radius ϵ and MinPts; finds arbitrary non-spherical clusters).
BIRCH	`Compress -> Cluster -> Refine` (Builds in-memory CF-Tree with $CF = (N, \mathbf{LS}, SS)$ in single scan of data).
Naïve Bayes	`Prior -> Conditional -> Multiply -> Compare` (Applies class-conditional independence).
Backpropagation	`Forward -> Error -> Backward -> Update` (Gradient descent backpropagation on multi-layer perceptrons).